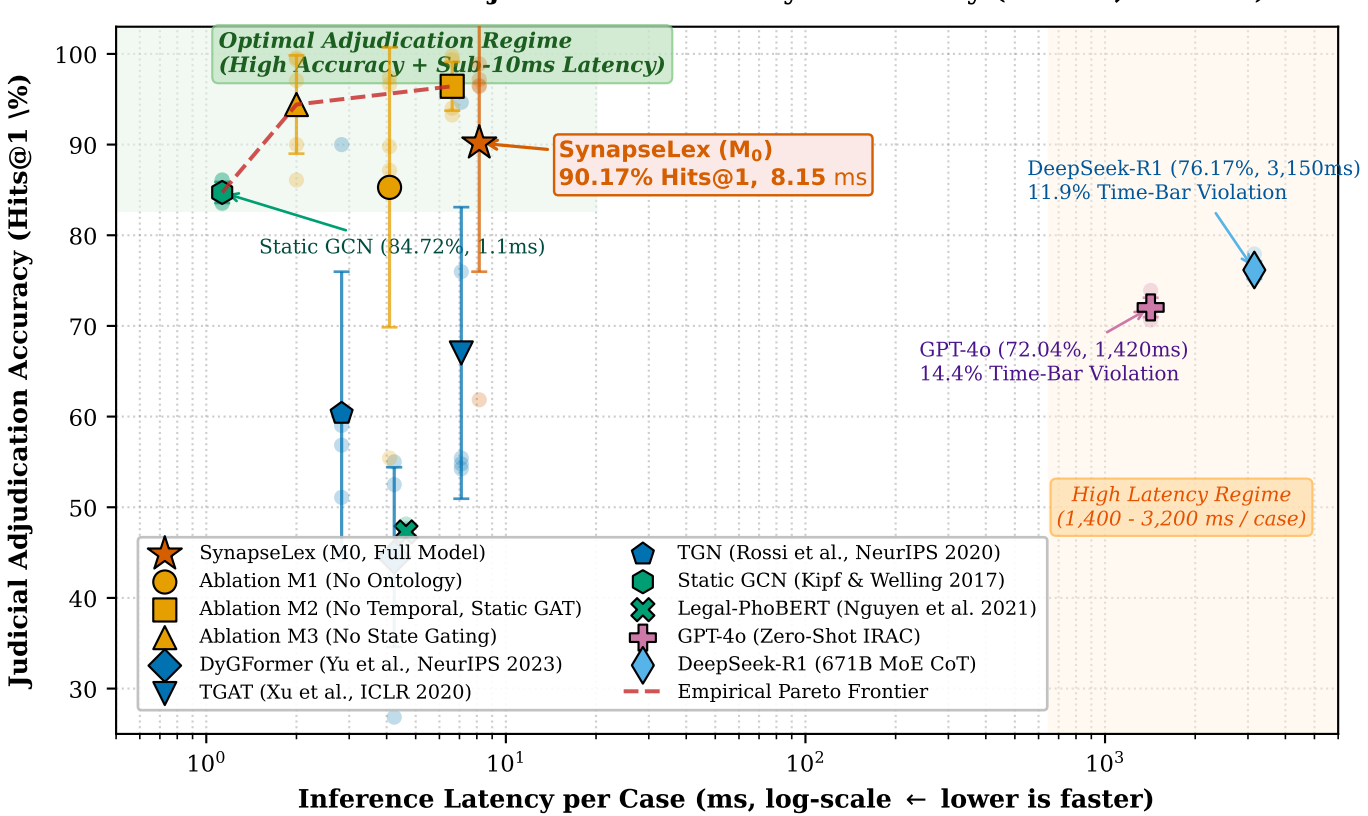

Pareto Frontier: Adjudication Accuracy vs. Latency ($N = 920$, 5 Seeds)
Optimal Adjudication Regime
(High Accuracy + Sub-10ms Latency)
SynapseLex (M_0)
90.17% Hits@1, 8.15 ms
Static GCN (84.72%, 1.1ms)
DeepSeek-R1 (76.17%, 3,150ms)
11.9% Time-Bar Violation
GPT-4o (72.04%, 1,420ms)
14.4% Time-Bar Violation
High Latency Regime
(1,400 - 3,200 ms / case)
SynapseLex (M0, Full Model)
Ablation M1 (No Ontology)
Ablation M2 (No Temporal, Static GAT)
Ablation M3 (No State Gating)
DyGFormer (Yu et al., NeurIPS 2023)
TGAT (Xu et al., ICLR 2020)
TGN (Rossi et al., NeurIPS 2020)
Static GCN (Kipf & Welling 2017)
Legal-PhoBERT (Nguyen et al. 2021)
GPT-4o (Zero-Shot IRAC)
DeepSeek-R1 (671B MoE CoT)
Empirical Pareto Frontier
Judicial Adjudication Accuracy (Hits@1 \%)
Inference Latency per Case (ms, log-scale ← lower is faster)
100
90
80
70
60
50
40
30
10^0
10^1
10^2
10^3